Department of Electronic Systems

Examination paper for TFE4146 Semiconductor Devices

Examination date: 27 May 2020

Examination time (from-to): 09:00 – 13:15

Permitted examination support material: All support material is allowed

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OTHER INFORMATION

If a question is unclear/vague – make your own assumptions and specify in your answer the premises you have made. Only contact academic contact in case of errors or insufficiencies in the question set.

Saving: Answers written in Inspera are automatically saved every 15 seconds. If you are working in another program remember to save your answer regularly.

Cheating/Plagiarism: The exam is an individual, independent work. Examination aids are permitted. All submitted answers will be subject to plagiarism control. *Read more about cheating and plagiarism here.*

Notifications: If there is a need to send a message to the candidates during the exam (e.g. if there is an error in the question set), this will be done by sending a notification in Inspera. A dialogue box will appear. You can re-read the notification by clicking the bell icon in the top right-hand corner of the screen. All candidates will also receive an SMS to ensure that nobody misses out on important information. Please keep your phone available during the exam.

Weighting: The weighting of the different problem sets (parts I to III) is specified for each set.

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How to digitize your sketches/calculations
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Remove personal information from the file(s) you want to upload

ABOUT SUBMISSION

Your answer will be submitted automatically when the examination time expires and the test closes, if you have answered at least one question. This will happen even if you do not click "Submit and return to dashboard" on the last page of the question set. You can reopen and edit your answer as long as the test is open. If no questions are answered by the time the examination time expires, your answer will not be submitted.

Withdrawing from the exam: If you wish to submit a blank test/withdraw from the exam, go to the menu in the top right-hand corner and click "Submit blank". This can not be undone, even if the test is still open.

Accessing your answer post-submission: You will find your answer in Archive when the examination time has expired.

Part I (weight 40%)

MULTIPLE CHOICE

There is only one correct answer to each of the questions 1 – 15 below. Correct answers are rewarded +2 points. Incorrect answers are awarded -1 point. [Note: It is OK to check two alternatives, but this of course implies a maximum score of +1 point for this question].

Question I – 1

Consider an electron moving in a semiconductor at room temperature under the influence of a small applied electric field. Which statement is correct?

a) The drift velocity is larger than the thermal velocity
b) The thermal velocity is larger than the drift velocity
c) The thermal velocity and the drift velocity are approximately equal

Question I – 2

The thermal equilibrium concentration of mobile charge carriers in Si at room temperature when doped with $N_D = 5\times10^{18}$ cm^{-3} (donors) and $N_A = 10^{18}$ cm^{-3} (acceptors) is,

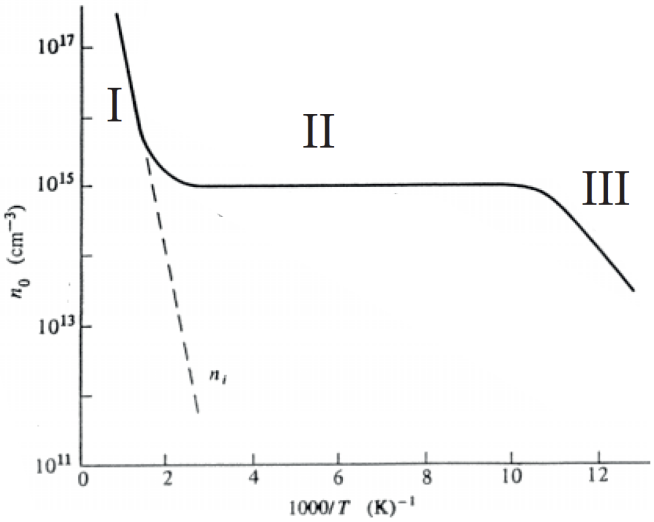
a) $n \approx 5\times10^{18}$ cm^{-3}, $p \approx 1\times10^{18}$ cm^{-3}
b) $n = p \approx \sqrt{5} \times 10^{18}$ cm^{-3}
c) $n \approx 4\times10^{18}$ cm^{-3}, $p \approx 0.5\times10^{2}$ cm^{-3}

Question I – 3

Under thermal equilibrium conditions in a p-n junction, the following current contributions always add up to zero,

a) The hole drift and diffusion currents
b) The electron and hole drift currents
c) The electron and hole diffusion currents

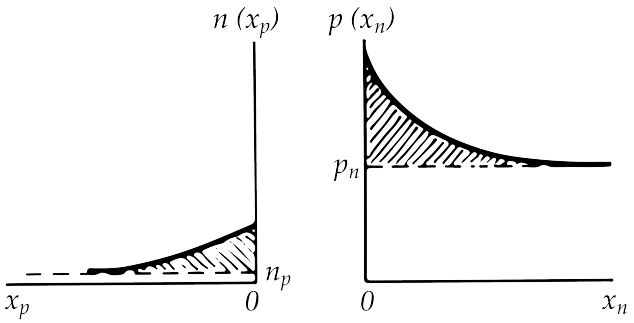
Question I – 4



The figure above traces the majority carrier (electron) concentration in thermal equilibrium vs. inverse temperature for an n-type Si substrate doped with 10^{15} donors/cm^3. A p$^+$-n junction fabricated on this substrate will not work for temperatures below approximately 100K (region III) because,

a) The thermal velocity of holes and electrons is insufficient for diffusion across the depletion zone
b) The temperature is too low for excitation of a sufficient number of electron-hole pairs
c) The temperature is insufficient for ionization of an appreciable fraction of the substrate donor atoms

Question I – 5



In a p-n junction, the carriers corresponding to the crosshatched regions in the figure above constitute a charge commonly referred to as:

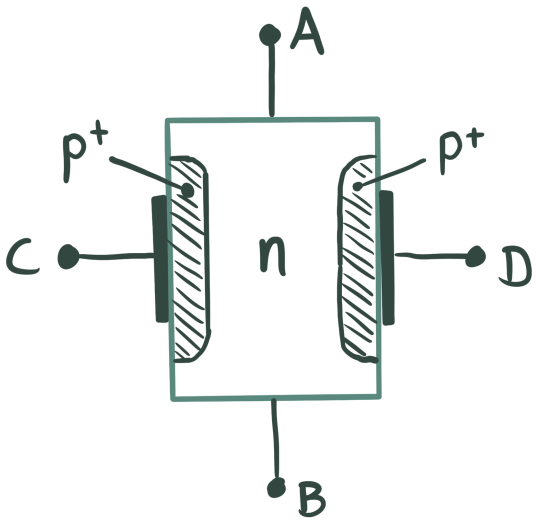
a) The *depletion* charge
b) The *stored* charge
c) The *inversion* charge

Question I – 6

Which combination of metal and semiconductor in an M/S contact is the best alternative for forming an *ohmic* contact?

a) n^+-doped poly-Si/n-type Si
b) n^+-doped poly-Si/p-type Si
c) Al/p-type Si

Question I - 7



The figure above depicts a uniformly n-doped slab of a semiconductor with ohmic contacts to the top and bottom faces (A,B) and abrupt p^+-n junctions with ohmic contact electrodes (C,D) to the p^+ regions. Which common semiconductor device does this figure represent if the terminals C and D are connected?

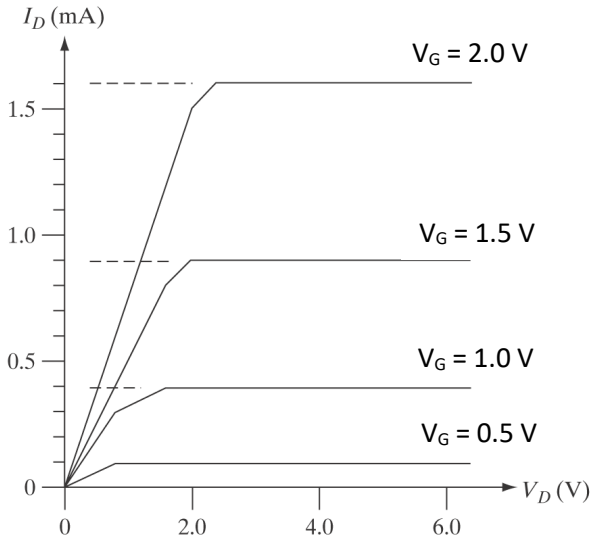
a) a diode
b) a JFET
c) a p^+-n-p^+ BJT

Question I – 8

Which is a true statement for an ideal Si MOSFET (MOS structure) at the onset of *strong inversion*?

a) The conductivity in the inversion channel is approximately the same as that of the bulk substrate
b) The intrinsic Fermi level in the inversion channel is equal to the semiconductor Fermi level
c) An inversion layer has just formed

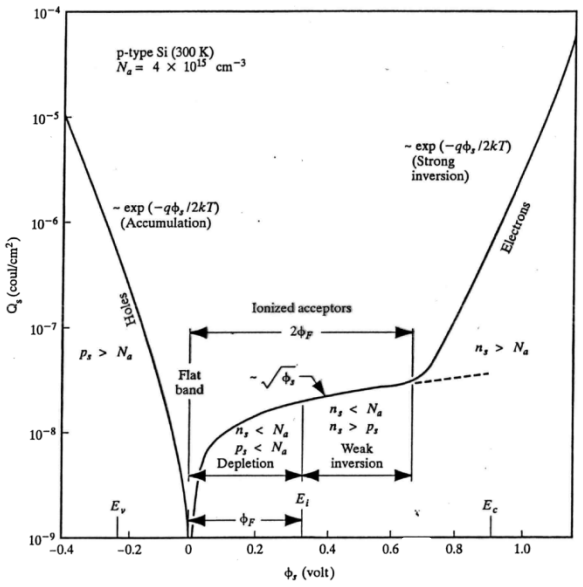
Question I – 9



The MOSFET I_D–V_D characteristics in the figure above were measured for,

a) A long-channel *enhancement mode* (i.e., normally OFF) *n-channel* MOSFET
b) A long-channel *enhancement mode* (i.e., normally OFF) *p-channel* MOSFET
c) A short-channel *enhancement mode* (i.e., normally ON) *n-channel* MOSFET

Question I – 10



The capacitance of an ideal MOS structure can be modeled as an insulator capacitance (C_i) in series with a voltage-dependent depletion capacitance (C_d). Assuming high-frequency operation (e.g., 1 MHz), under which circumstances will the overall capacitance be dominated by C_i (i.e., $C_{tot} \approx C_i$)? [*Hint:* cf. the plot of surface charge Q_s vs. surface potential ϕ_s above].

a) Depletion and strong inversion
b) Accumulation and strong inversion
c) Accumulation and depletion

Question I – 11

Which quality should the *subthreshold slope*, S, preferably have for a MOSFET to be used as a switch?

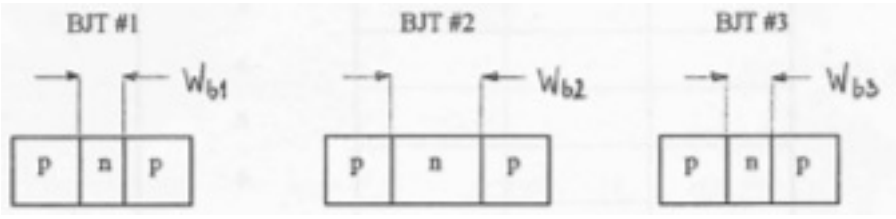
a) S should be small, as this will only require a small gate voltage swing for modulation of the drain current
b) S should be large, as this implies a small interface capacitance, which is important for high frequency operation
c) S should be close to 0.7 V, as this ensures that thermal fluctuations do not open and close the channel.

Question I - 12

The *subthreshold slope*, S, of a MOSFET [mV/decade] is improved by,

a) Decreasing the oxide thickness and increasing the substrate doping
b) Decreasing the oxide thickness and decreasing the substrate doping
c) Increasing the oxide thickness and decreasing the substrate doping

Question I – 13



The three p-n-p bipolar transistors in the figure above are all symmetric and identical, except for BJT #2 having twice the base width of BJT #1 ($W_{b2}=2\times W_{b1}$), and BJT #3 having twice the base doping of BJT #1 ($N_{B3}=2\times N_{B1}$). The emitter efficiency γ can be expressed on the form,

$$\gamma = \left[1 + \frac{L_p^n n_n \mu_n^p}{L_n^p p_p \mu_p^n} tanh\left(\frac{W_b}{L_p^n}\right)\right]^{-1}$$

where superscripts denote which side of the emitter-base junction the diffusion lengths $L_{n,p}$ and mobilities $\mu_{n,p}$ are referred to. [*Hint:* The base width W_b may be assumed << L_p^n, the hole diffusion length].

Which of these three BJTs have the highest value for the emitter efficiency, γ ?

a) BJT #1
b) BJT #2
c) BJT #3

Question I – 14

Decreasing the base width of a bipolar junction transistor will (assuming the device remains unchanged otherwise),

a) Increase the magnitude of the Early voltage, V_A
b) Leave the magnitude of the Early voltage, V_A , unchanged
c) Decrease the magnitude of the Early voltage, V_A

Question I – 15

A Si photovoltaic solar cell with *fill factor FF = 0.8* delivers a maximum power of 28 mW under full illumination. The open-circuit voltage V_{oc} and short-circuit current I_{sc} of this PV cell is,

a) V_{oc} = 1.4 V and I_{sc} = 25 mA
b) V_{oc} = 1.0 V and I_{sc} = 28 mA
c) V_{oc} = 0.7 V and I_{sc} = 50 mA

Part II (weight 30%)

Question II - 1

Consider an n^+-p-n Si bipolar junction transistor (BJT) with dopant concentrations $N_E > N_B > N_C$.

Fill in the blanks (numbered) in the following section of *italicized* text. Words to insert are suggested for each blank/number. Choose the appropriate alternative.
[Answer on the form; (1) – (*correct word*), (2) – (*correct word*), etc.]

When biased in the normal active mode, the collector current in this BJT is predominated by (1) traversing the neutral base region (B) without recombination. The base current i_B *flows (2) B and controls an amplified current in the collector circuit when the BJT is connected in the (3) configuration. The amplification is given approximately as* $\beta \approx$ *(4), where* τ_n *is the recombination lifetime for (5) in B, and* τ_t *is the average transit time for (6) across B. When the collector terminal is short-circuited to the base terminal, the excess hole concentration at the collector end of B is* Δp_C *= (7). When used for switching, the following 3 operation regimes are involved – the (8) mode. If the p-type base region is ion-implanted so that the dopant concentration* N_B *decreases from emitter towards collector, the carrier transport across the base takes place by (9). Emitter crowding and Early effect in a BJT result from (10), respectively.*

(1) electrons, holes
(2) into, out of
(3) common-base, common-emitter
(4) τ_t/τ_n, τ_n/τ_t, $1-\tau_t/\tau_n$, $1-\tau_n/\tau_t$
(5) holes, electrons
(6) electrons, holes
(7) $-n_{p0}$, p_{n0}, 0, n_{p0}
(8) normal active, inverse active, and saturation **;** normal active, cutoff, and inverse active **;** normal active, cutoff, and saturation
(9) diffusion of electrons, drift of electrons, both drift and diffusion of electrons
(10) increased emitter efficiency and reduced effective base width **;** ohmic losses in the base and reduced effective base width **;** increased emitter efficiency and avalanche multiplication in the base

Question II - 2

The text below includes two paragraphs taken from a column in the Norwegian magazine *Teknisk Ukeblad* (*Technical Weekly*) with the title "*Fra lys til strøm*" (from light to power) some years ago. Two paragraphs of this column, named "*Den fotoelektriske effekt*" (the photoelectric effect) and "*Lyset driver effekten*" (the light drives the power) describe the fabrication and operation principle of a photovoltaic (PV) solar cell.

- Comment on this description of the fabrication and operation principle of PV solar cells with special emphasis on mistakes/misconceptions in this account.

The photoelectric effect

The photoelectric effect can be found in all sorts of materials. Light quanta (photons) kick out electrons bound to the atoms in the material. In a solar cell, the free electrons are "forced" to pass through a circuit before they drop back in place. In that way, their electric energy can be utilized in the same way as electricity in general.

This is accomplished by placing two thin sheets of silicon with slightly different physical properties against one another. The conductivity of silicon is controlled by adding minor quantities of a different material. This is referred to as doping of the semiconductor.

One sheet is doped with the element boron, which has one electron less than silicon in the outer shell. This doping makes p-type silicon, which is positively charged. The other sheet is doped with phosphor, which has one electron more than silicon in the outer shell. Such doping makes n-type silicon, which is negatively charged. When these sheets are brought in contact, electrons will jump from the n-side to the p-side. Since the electrons are negatively charged, the current will pass in the opposite direction. A thin layer is formed, a p/n junction, with a static electric potential across the interface between the two sheets.

The light drives the power

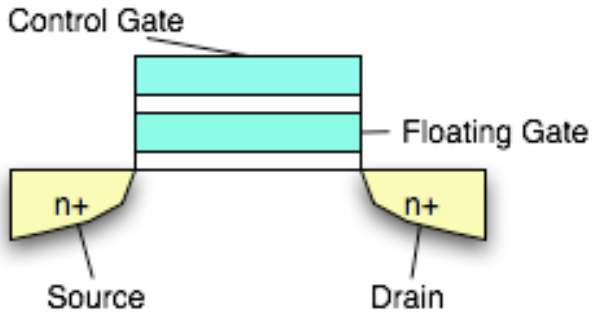
The built-in potential cannot generate a net current in an outer circuit, since positive and negative charge keep one another at bay. This is where the light comes in and disturbs the balance.

The energy in the sunlight is sufficient to kick electrons residing in the vicinity of the interface layer across the potential barrier present there (the so-called bandgap, which is 1.1 electron volt in silicon), so that an electron-hole pair is formed on each side of the barrier.

These electrons figure out that the simplest route back in place is to pass through the outer circuit, where their electrical power can be utilized. The outer voltage of a silicon cell is approximately 0.5 volt, independent of the surface area, while the current through the outer circuit is proportional to the amount of solar radiation which hits the cell.

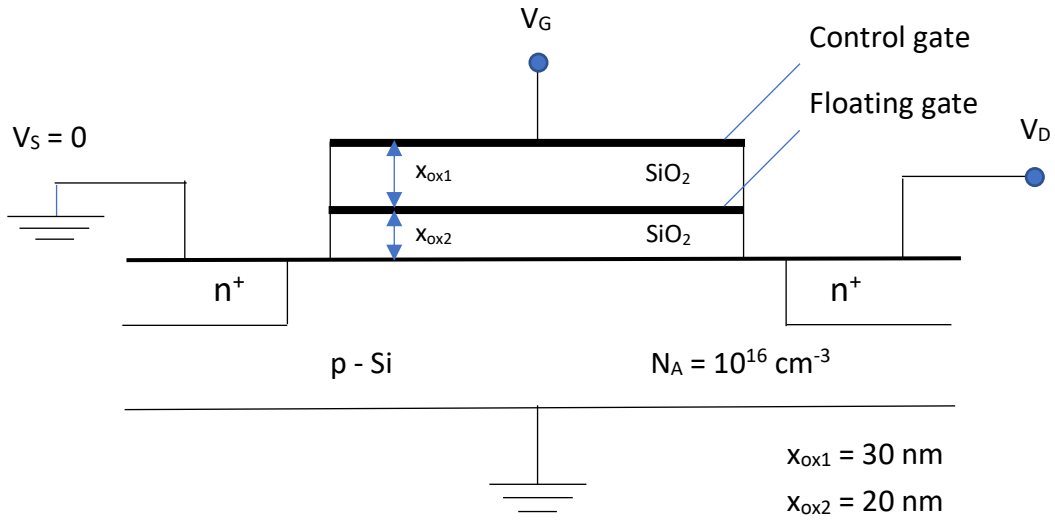
Part III (weight 30%)

MOSFET – EPROM



The **erasable programmable read-only memory** (EPROM) is a type of programmable memory chip that retains the data when the power is switched off. The EPROM circuit has an array of *floating-gate* (MOSFET) transistors, illustrated in the figure above. The *floating-gate* electrode has no connection to other parts of the integrated circuit. Another layer of oxide is deposited on top of the *floating gate*, before deposition of the *control gate*. Both gates are usually made from n^+-doped poly-Si.

The floating-gate electrode is very thin, so that we may consider a stored charge on this electrode as a surface (2D) charge, separated a distance x_{ox2} from the SiO_2/p-Si substrate interface, (cf. the figure below).



We assume identical work functions, $\Phi_{poly\text{-}Si} = \Phi_{Si}$, for poly-Si and single-crystalline Si material of identical type (n,p) and the intrinsic Fermi level (E_i) in the center of the bandgap ($E_i = E_{VBM} + E_g/2$). Moreover, we neglect surface charges (Q_i) at the SiO_2/Si interface.

- Find the threshold voltage V_T for this device with no charge deposited on the *floating-gate* electrode. [The total oxide thickness for the *control-gate* electrode is then $x_{ox1}+x_{ox2}$].

  Numerical answer required!

- Determine the sign and numerical value of the charge [pr. unit area] required on the *floating-gate* electrode in order to shift the threshold voltage V_T up by 8 V, [i.e., $\Delta V_T = +8V$].

  Hint: Use Gauss' law to establish the potential V_1 and V_2 across the two oxide layers of the gate contact stack, x_{ox1} and x_{ox2}, respectively.

The drain current characteristics (I_D–V_D) of the transistor in the saturation regime ($V_D > V_{sat}$) can be expressed on the form,

$$I_D \approx \frac{k}{2}(V_G - V_T)^2 \qquad \text{where} \qquad k = \frac{Z}{L} \times \mu_n C_i$$

- Use this drain current characteristics to determine I_D for a gate voltage $V_G = 3V$ under the conditions stated in the two bullet points above, i.e., with and without charge on the *floating-gate* electrode. Assume a value for $k = 50\ \mu A/V^2$.

Information given:	relative dielectric permittivity of Si	$\varepsilon_r(Si) = 11.8$
	relative dielectric permittivity of SiO_2	$\varepsilon_r(SiO_2) = 3.9$
	permittivity of free space	$\varepsilon_0 = 8.85\times10^{-14}$ F/cm^2
	intrinsic carrier density of Si	$n_i = 1.5\times10^{10}$ cm^{-3}
	Si bandgap	$E_g = 1.11$ eV